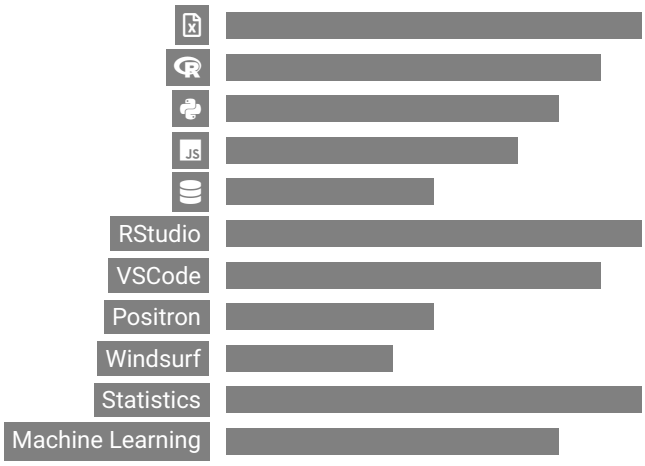
TINGWEI ADECK

Data Scientist

Duncanville, TX
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GitHub
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ORCIDiD

EXECUTIVE SUMMARY

Ivy league trained Data Scientist and Tech Enthusiast, with **3+** years of experience building data software solutions in languages like R and Python. Proven open-source developer of **2** R packages (normfluodbf and tidyDenovix), with **135k+** downloads combined. Developer of **2** Python packages (faced and arrimputer) hosted on PyPi.
Served as an Energy Transition Data Science Intern at the University of Houston, where I applied statistical modeling, geospatial analysis, and machine learning to support research on clean energy adoption, grid modernization, and decarbonization strategies.
Currently positioned to help organizations in solving data related problems. Let's connect ASAP, to discuss how we can collaborate.



EXPERIENCE

UNIVERSITY OF HOUSTON

DATA SCIENCE ENERGY TRANSTION INTERN

Houston, Texas

2025 – 2026

· Developed an **R Shiny GUI app** for data ingestion and variable type manipulation, saving users **1+ minutes** per operational task via **code-writing avoidance**.
· Utilized **machine learning clustering algorithms**, in grouping wells by **mineral composition**, based on merged **X-ray diffraction** studies and **thermal conductivity** data.
· Built **geospatial visualizations** using **QGIS** by integrating satellite and **Alternative Energy Fuel Stations** data to map energy demand patterns across residential, commercial, and industrial zones.

SELF-EMPLOYED

DATA SCIENCE FREELANCER

Shawnee, KS

2022 – 2024

· Developed 2 R packages that generated several users from the biomedical field.
· Extended my skillset from R programming, into python programming by developing 2 Python packages in the field of image processing and machine learning.
· Prepared my profile for Public Health, Biostatistics or Data Science programs. Accepted to study Biostatistics, by Columbia and Brown University.

UNIVERSITY OF MISSOURI KANSAS CITY

GRADUATE RESEARCH ASSISTANT

Kansas City, Missouri

2020 – 2022

· Performed **PhD** research on **Liposome Flux Assays (LFAs)** (high throughput system for **drug screening**), in the lab of **Dr. Wang**.
· Developed **research protocols** based on literature reviews. Implementation of the protocols proved sound understanding of research literature and experimental concepts.
· Performed large-scale **wet lab experiments**, used to obtain scientific biological data for further analysis. Followed up data collection, with **data analysis** and **dissemination** of results to PIs and peers, in **lab meetings** and **conferences**.

EDUCATION

BROWN UNIVERSITY

SCM BIOSTATISTICS HEALTH DATA SCIENCE

Providence, Rhode Island

2025 – Now

· **Relevant Coursework:** Probability, Statistics, Machine Learning and Artificial Intelligence.
· **Capstone: Sensitivity Analysis** on the **Logistic Regression Model** of the Relationship Between Chronic Obstructive Pulmonary Disease (COPD) and Psychological Distress in Farmers.

UNIVERSITY OF MISSOURI KANSAS CITY

BIOLOGY/BIOINFORMATICS

Kansas City, Missouri

2022

· **Relevant Coursework:** Introduction to SAS, Introduction to SQL, Applied Biostatistics I, Advanced Experimental Molecular Biology I and II.
· **PhD Project:** Understanding **Liposome Flux Assays (LFAs)** in the context of **Sodium Voltage-gated Channels (NavAb)**.

WEST TEXAS A & M UNIVERSITY

MBA/BIOLOGY

Canyon, Texas

2015 – 2018

· **Relevant Coursework:** Econometrics, Bioinformatics, Introduction to Biometry, Analytical Chemistry, Calculus I and II.
· **Project:** Investigating the role of **Tryptophan-Hydroxylase I** as a mediator in the **circadian** regulation of **Serotonin** in **mice gut**.

PROJECTS

Normfluodbf

- Developed an open-source R package and interactive Shiny application optimized for the end-to-end processing and visualization of liposome flux assay data.
- The package provides a modular data analysis pipeline for importing data, analyzing data and visualizing data.

TidyDenovix

- Authored a specialized R package engineered to streamline data extraction, quality control, and downstream analytics for the Denovix DS spectrophotometer family of products.
- Architected the package framework on the backbone of the tidyVerse, unlike previous work built with over-reliance on in-house code.

RandVar

- An educational **R Shiny App** for the exploration of **statistical distribution** types. Built during the first semester at Brown, as a tool to teach **statistical distributions**.
- Implementation of GitHub Collaboration with Brown coursemates, in designing and implementing the application.

RELEVANT COURSES

Probability and Statistical Inferencing - Dr. Ishaan Dave, Ph.D (Brown)
Statistical Programming - Dr. Beverly Gonzalez, MS, ScM, Ph.D (Brown)
Linear and Generalized Linear Models - Dr. Crystal Linkletter, Ph.D (Brown)
Causal Inference - Dr. Peter Lipman, Ph.D (Brown)
Statistical Machine Learning - Dr. Tao Liu, Ph.D (Brown)

LANGUAGES

English - native
French - native